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Surface-Based Predictive Monitoring for Electric Submersible Pumps Using Electrical Signature Analysis

G. ten Hoeve, M. Otero, and S. F. W. Jagers, Samotics, Leiden, Netherlands; B. Haapanen, R. Dodson, and B. Hicks, ChampionX, Houston, Texas, United States

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Abstract

Electric Submersible Pumps (ESPs) are a critical technology for artificial lift in the oil and gas industry, utilized in an estimated 150,000-200,000 wells worldwide. The high cost of workovers, ranging from USD 50,000 to USD 100,000 per day for heavy-duty land rigs, makes early detection of ESP degradation invaluable for operators. Conventional monitoring methods rely on downhole gauges, which are costly, have slow data update rates (approximately 30-second data samples), and exhibit a five-year survival rate of less than 65% in field studies. This paper introduces SAM4, a continuous online surface-level Electrical Signature Analysis (ESA) system that extracts health and efficiency data from current and voltage signals, with outbound data traffic capped at approximately 2 GB per ESP per month. A 20-asset pilot project in the Permian Basin demonstrated the system's ability to provide early detection of electrical, mechanical, and operational faults without well intervention. This paper details the project's assumptions, technical adaptations, field results, current status, and a future roadmap that includes a cloudless edge version, positioning ESA as a key cost-effective monitoring tool for ESPs globally.

Introduction and Background

The global oil and gas industry operates under significant economic and operational pressures, with a constant drive to enhance efficiency and reduce production costs. Electric Submersible Pumps (ESPs) are a cornerstone of artificial lift technology, essential for maintaining production in a vast number of wells globally ([Fortune Business Insights, 2023](#)). The global electric submersible pump market was valued at USD 10.4 billion in 2023 and is projected to reach USD 19.7 billion by 2033, growing at a CAGR of 6.5% from 2024 to 2033 ([Allied Market Research, 2024](#)). However, the operational integrity of ESPs is a major concern for producers. ESP downtime not only defers production but also necessitates costly workovers, with day rates for 3,000 HP land rigs often exceeding USD 50,000 per day ([Westwood, n.d.](#)).

Traditionally, monitoring ESPs has relied on downhole gauges, which, while providing direct measurements, have several limitations. These sensors offer slow data sampling rates, typically around 20-30 second intervals, which is often insufficient for real-time diagnostics or advanced control algorithms. Furthermore, downhole gauges add a significant cost to well completions, estimated at USD 6,000-15,000

per installation. Their reliability is also a concern; when these gauges fail, operators lose visibility into the well's operation, often leading to a run-to-failure approach that results in high overhaul costs and deferred production.

This creates a significant monitoring gap and an opportunity for more advanced, surface-based solutions. Electrical Signature Analysis (ESA) has emerged as a powerful alternative. The SAM4 system, for instance, has demonstrated over 90% effectiveness in detecting faults in centrifugal pumps across more than 10,000 surface and submerged assets (internal data). Given that ESPs are essentially stacked centrifugal pumps, there is a strong theoretical basis for the successful application of ESA to this critical asset class. The primary goal of this project is to demonstrate that surface-based ESA can meet or exceed the performance of downhole sensing while substantially reducing monitoring costs, thereby offering a more sustainable and economically viable solution for the industry.

Assumptions and Project Objectives

The successful application of ESA to ESPs is predicated on three key assumptions:

1. **Fault Mode Similarity:** It is assumed that the primary fault modes in ESPs are analogous to those observed in conventional centrifugal pumps. This assumption is validated by matching known ESA fault signatures with ESP-specific failure scenarios.
2. **Signal Integrity:** The project assumes that the long downhole cables and surface-level transformers do not excessively distort the electrical signals, thereby preserving the integrity of the ESA data for accurate diagnostics. This is validated through comparative spectral analysis of the signals.
3. **Economic Viability:** A core assumption is that surface-based ESA monitoring is an order of magnitude more cost-effective than traditional downhole gauges. This is validated through a comprehensive Capital Expenditure (Capex) and Operational Expenditure (Opex) analysis, demonstrating a significant return on investment (ROI).

The technical objectives of the project are to demonstrate the accuracy of fault detection from surface-level measurements, while the commercial objective is to provide a compelling and economically attractive monitoring alternative for wells that currently lack traditional gauges.

Table 1—Key Project Assumptions and Validation Targets

Category	Assumption	Validation Target
Failure physics	ESPs share dominant fault modes with surface centrifugal pumps and/or submersed (multistage) pumps, like borehole pumps in the clean water industry.	Replicate detection patterns.
Signal fidelity	3 km cables + step-up transformers do not mask ESA harmonics. Outperforms downhole gauge on data quality and sampling rate to enable predictive capabilities.	Compare pre/post spectra.
Economics	Surface ESA must cost an order of magnitude less than downhole gauges + rig. In addition to reducing costs, it should unlock significant upside potential by enabling earlier interventions, production optimization, and longer ESP run life.	Capex/Opex model; strong ROI from both avoided costs and additional production gains.

Technical Solution: ESA for ESPs

The SAM4 system leverages Electrical Signature Analysis (ESA) to monitor Electric Submersible Pumps (ESPs) from the surface, offering a non-invasive and highly effective diagnostic solution. The implementation of this technical solution involves specific hardware configurations, advanced signal

processing techniques, comprehensive fault classification capabilities, and sophisticated self-adaptation mechanisms.

Hardware and Configuration

The physical setup of the SAM4 ESA system for ESP monitoring is designed for ease of installation and optimal signal capture. The primary components, clamp-on current transformers (CTs) and voltage transformers (VTs), are strategically mounted downstream of the step-up transformer. This placement is crucial as it allows the system to capture the electrical signals after they have been conditioned for the ESP motor, providing a direct reflection of the power delivered to and consumed by the downhole pump. By clamping onto existing cabling, the installation process is non-intrusive, avoiding the need to cut or modify the power lines, which is a significant advantage in operational environments.

Following the CTs and VTs, the monitoring device and gateway are securely placed inside the Motor Control Cabinet (MCC). The MCC typically houses the variable frequency drive (VFD) and other control electronics for the ESP, making it a convenient and secure location for the ESA hardware. This integrated setup minimizes the footprint of the monitoring equipment on site.

A critical feature of the system is its edge unit, which is capable of streaming high-resolution data snippets at 64 kS/s (kilo-samples per second) to the cloud. This high sampling rate is fundamental for capturing the intricate harmonic content of the electrical signals necessary for detailed ESA diagnostics, distinguishing it significantly from the slow update rates of downhole gauges. For data transmission, the system offers versatile connectivity options, supporting 4G, Wi-Fi, or Ethernet connections. During the initial field trial, a 4G connection was successfully utilised, demonstrating its reliability in typical oilfield environments.

A cornerstone of the SAM4 architecture is its bandwidth-light design. This innovative approach ensures that even with high-resolution data acquisition, the system manages to cap outbound traffic at approximately 2 GB per ESP per month. This low data consumption is achieved through the transmission of compressed waveforms and delta trends, rather than raw, continuous streams of data. The significance of this low bandwidth requirement cannot be overstated, particularly for remote leases where high-speed internet is scarce or prohibitively expensive. Keeping cellular plans modest through this efficient data management directly contributes to the system's cost-effectiveness and broad applicability in challenging operational terrains.

Signal Processing

The efficacy of the ESA system hinges on sophisticated signal processing capabilities. A notable technical adaptation for ESP monitoring is the implementation of an algorithm specifically designed to effectively monitor ungrounded voltages. ESP power systems often operate with ungrounded or high-resistance grounded configurations, which can present unique challenges for traditional voltage monitoring. This specialised algorithm ensures that accurate and reliable voltage data can be extracted under these conditions, which is crucial for identifying certain types of electrical faults. Importantly, despite these specific adaptations for ESPs, the core AI models that drive the diagnostic capabilities of SAM4 remain largely unchanged. This consistency allows the system to leverage its extensive, proven fault detection logic developed for other pump types, streamlining development and ensuring robust performance.

Fault Classes

The SAM4 ESA system is engineered to detect a comprehensive range of fault types, categorised into three primary classes ([MDPI, 2023](#)):

- **Electrical faults:** These encompass a variety of issues related to the electrical supply and components of the ESP system. Examples include imbalance in the three-phase power supply, harmonic distortion indicating nonlinear loads or power quality issues, rotor-bar damage within the motor, general power quality excursions, cable damage along the long power lines, and impending

ground faults in either the cable or the motor itself. Early detection of these electrical anomalies is critical for preventing catastrophic motor failure and extensive cable damage.

- **Mechanical faults:** These refer to physical degradation within the pump or motor assembly. The system can identify indicators of rotor eccentricity (where the rotor is off-centre), misalignment between pump and motor sections, bearing wear within the motor or pump stages, and impeller erosion within the pump stages. Mechanical failures often manifest as distinct vibration patterns that are translated into subtle electrical signature changes, allowing for surface detection.
- **Operational faults:** These relate to suboptimal operating conditions or external process issues that impact the ESP's performance. Detectable operational faults include gas lock (where gas accumulates and impedes fluid flow), dry-run conditions (operation without sufficient fluid), excessive cycling (frequent starting and stopping), and off-BEP (Best Efficiency Point) loading. These issues can significantly reduce pump efficiency, accelerate wear, and lead to premature failure, making their early detection vital for process optimisation and extended run life.

Self-Adaptation

A key feature enhancing the reliability and accuracy of the SAM4 system is its self-adaptation capability. After an initial approximately 72 hours of continuous operation, the monitoring system is capable of establishing a baseline of normal operation for the specific ESP it is monitoring. This learning period allows the system to characterise the unique electrical signature of the pump under various operating conditions. Crucially, the system is designed to compensate for Variable Frequency Drive (VFD) speed changes and to adapt to different motor types, whether Permanent Magnet Motors (PMM) or induction motors.

This adaptive logic is vital because ESPs, particularly in unconventional wells, often experience significant and frequent load swings and operate across a wide range of speeds and loading situations. The system intelligently learns how to adapt to these dynamic changes, allowing it to effectively extract the specific voltage and current signal features that are indicative of faults, rather than merely reflecting normal operational variations. This intelligent self-calibration ensures that the diagnostic models remain accurate and sensitive to true anomalies, preventing false positives and providing reliable insights into the ESP's health, regardless of its control mode or motor design.

Field Deployment and Results

A comprehensive field trial involving 20 ESPs in the Permian Basin over eight months validated the system's capabilities.

Key Challenges

Several key challenges were addressed during the field deployment:

- **Sensor Installation:** Initial delays in vendor scheduling were mitigated by developing pre-configured kits that simplify installation and power delivery using Power over Ethernet (PoE) technology, attached to the existing surface-level electrical equipment (MCC).
- **Cloud Dependence:** Two leases with weak LTE/SATCOM connectivity experienced upload lags, a common issue in remote IIoT operations ([IIoT World, 2025](#)). This was addressed by the system's low data traffic (≤ 2 GB/month/ESP), which allows for the use of low-cost satellite solutions like Starlink. On-device buffering ensures data is retained until a connection is re-established. A cloudless edge mode is also on the roadmap.
- **Process Variability:** The significant load swings characteristic of ESPs in unconventional wells were addressed by implementing an adaptive baseline logic, which is now a default feature of the system.

Table 2—Key Field Deployment Challenges and Mitigations

Challenge	Effect	Mitigation
Sensor installation	Vendor scheduling delays on two wells.	Pre-configured kits that are simple to attach and power using PoE (Power over Ethernet) technology to the existing surface-level electrical equipment (MCC) (see § 6-1).
Cloud dependence	Two leases with weak LTE/SATCOM suffered upload lags; similar IIoT connectivity issues are reported for remote operations (IIoT World).	Low traffic (≤ 2 GB/month/ESP) kept data costs down, allowing the use of a low-cost satellite solution (e.g., Starlink), with on-device buffering that held data until the signal returned. A cloudless edge mode is on the roadmap (§6-4).
Process variability	ESP load swings (in unconventional) exceed those of surface pumps.	Adaptive baseline logic (now default).

Detection Highlights

- **Sine-Wave-Filter Failure:** A significant spike in harmonics was flagged four days before a VFD trip, leading to the timely replacement of the filter and preventing downtime.
- **Failed Cable Plug:** Multiple voltage indicator spikes, including negative sequence voltage, confirmed the failure detection model's accuracy.
- **Mechanical Unbalance:** A rise in energy at 2x and 4x the rotational frequency indicated mechanical unbalance, which was mitigated by changing the operational mode, preventing further degradation.
- **Gas Locking:** Power sags and jumps in rotor-eccentricity indicators in two ESPs were correlated with SCADA flow loss data, confirming gas locking events.

Quantified Benefit

Sine Wave Filter Failure. The SAM4 system detected a severe increase and unbalance in the main harmonics, specifically:

- **Current phase mismatch:** Phases 1 and 3 showed significant harmonic distortion, while Phase 2 remained relatively clean.
- **Unusual harmonics:** The presence of even harmonics (e.g., 2X, 4X, 6X) was detected, which is atypical for balanced three-phase systems.
- **Voltage Imbalance:** Detected discrepancies pointed to potential issues in the drive train, likely post-VFD.

ChampionX, working closely with Samotics, initiated a targeted diagnostic approach:

1. **Powertrain segmentation:** Analyzed electrical signatures at each key segment: the VFD output, the sine wave filter output, and the step-up transformer VT output.
2. **Hypothesis validation:**
 - Checked for VFD modulation faults like dead-time distortion.
 - Investigated the sine wave filter for asymmetrical component degradation.
3. **Harmonic comparison:** Benchmarked this ESPs voltage and current harmonics against similar ESPs. This confirmed the issue was isolated to this particular ESP and not systemic.

Through harmonic analysis and comparative data, the team narrowed the fault to a likely degradation in the sine wave filter components, which were no longer symmetrically conditioning the VFD output.

ChampionX engineers were able to repair the filter in a timely manner, preventing the supply of unfiltered pulse width modulation (PWM) waveforms to the ESP. Unfiltered PWM waveforms can cause premature insulation failure in the cable or motor and excessive heating, potentially leading to internal ‘fire’ and motor failure. This alert and the subsequent intervention reduced the risk of early failure and extended the lifetime of the ESP.

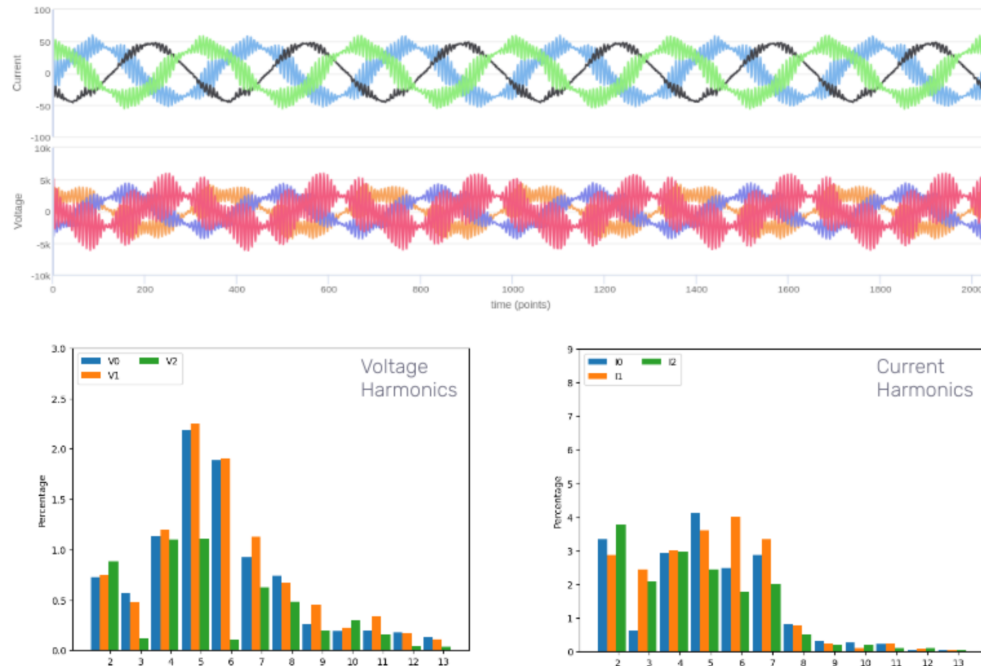


Image 1—Top: ESP system with a failed sine-wave filter - Current and voltage 3-phase waveforms (0.1 seconds of 20 kHz data sample) measured on the MV side of the step-up transformer; Bottom: Details of the harmonic distortion content per phase (up to the 13th harmonic).

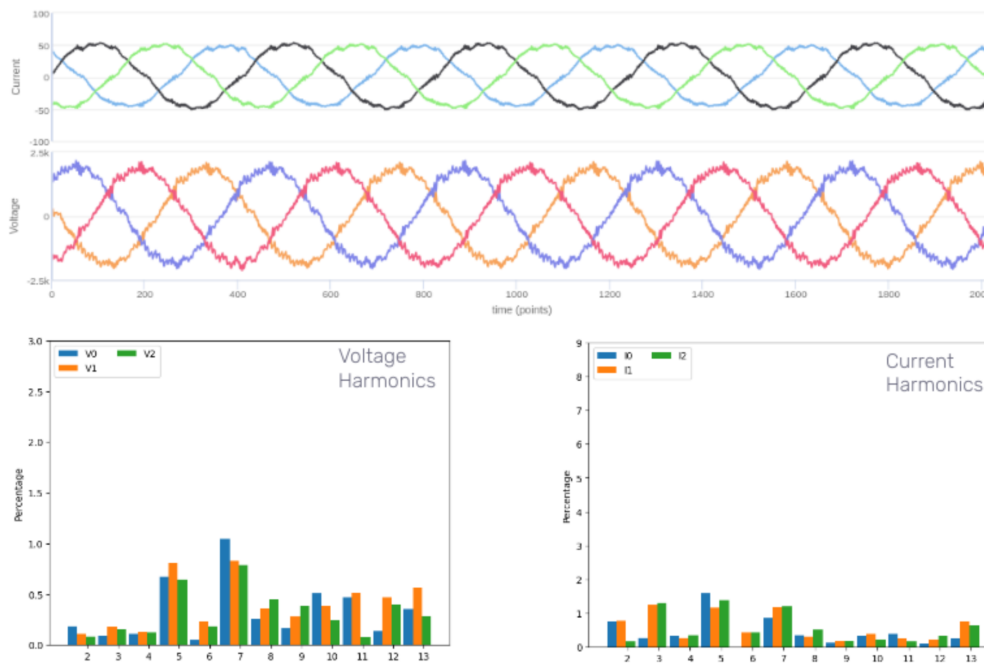


Image 2—ESP system with a healthy sine-wave filter - Current and voltage 3-phase waveforms (0.1 seconds of 20 kHz data sample) measured on the MV side of the step-up transformer; Bottom: Details of the harmonic distortion content per phase (up to the 13th harmonic).

Failed Cable Plug (Failure Point in Motor Pothead). Following a reported ESP failure, a failed motor pothead cable plug was identified. A thorough review of the ESA data revealed clear indicators in the voltage signal that predicted the impending failure. These indicators, which track voltage unbalance and negative sequence, are normalized for ESP operational fluctuations. This analysis has led to the development of new detection models specifically for this type of ESP failure, enabling precise, 24/7 predictive detection of cable plug degradation.

The ability to detect early indicators of cable plug degradation can help operators make better operational decisions before a failure occurs. If these warning signals are identified early enough, they can guide actions such as adjusting pump speed or load to reduce electrical stress on the cable, potentially extending its run life. Alternatively, if the signals indicate that failure is imminent and unavoidable, operators can proactively plan a workover to avoid unexpected downtime. Over time, these insights could allow operators to move from simply reacting to cable plug failures to actively managing pump operation and intervention timing to minimize their impact on production.

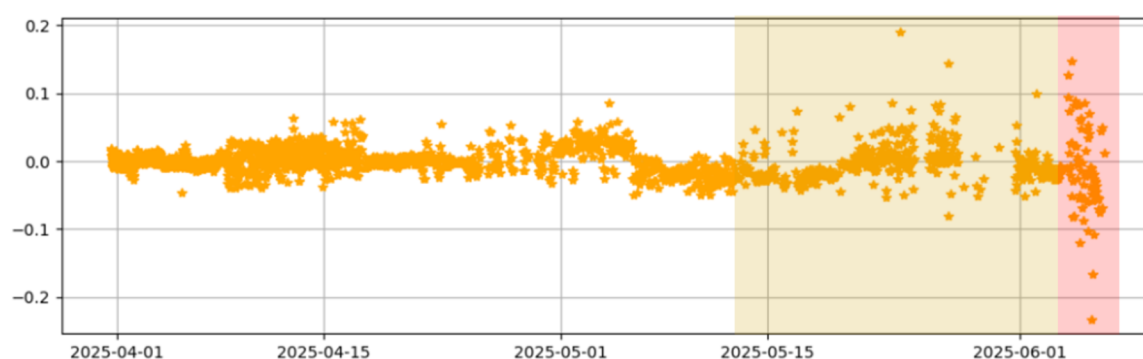


Image 3—ESA voltage unbalance indicator tracked over time

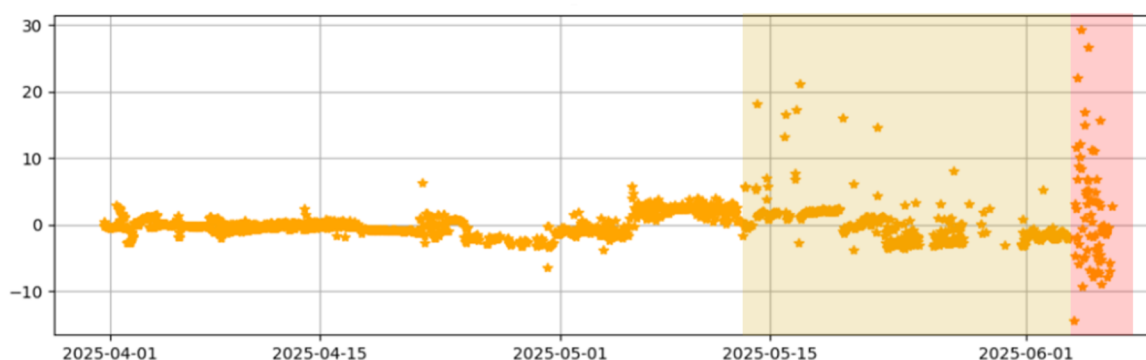


Image 4—ESA voltage negative sequence indicator tracked over time

Gas Locking. The SAM4 system identified gas-locking through a combination of excessive cycling, inconsistent active power consumption and increases in the rotational frequency component (mixed rotor eccentricity), indicating unexpected changes in the process or pumped medium. ChampionX was able to confirm the gas-locking condition and ensure the situation remained within safe operating bounds. After a restart and activation of the gas-lock recovery mode, the ESP gas-locking situation was resolved.

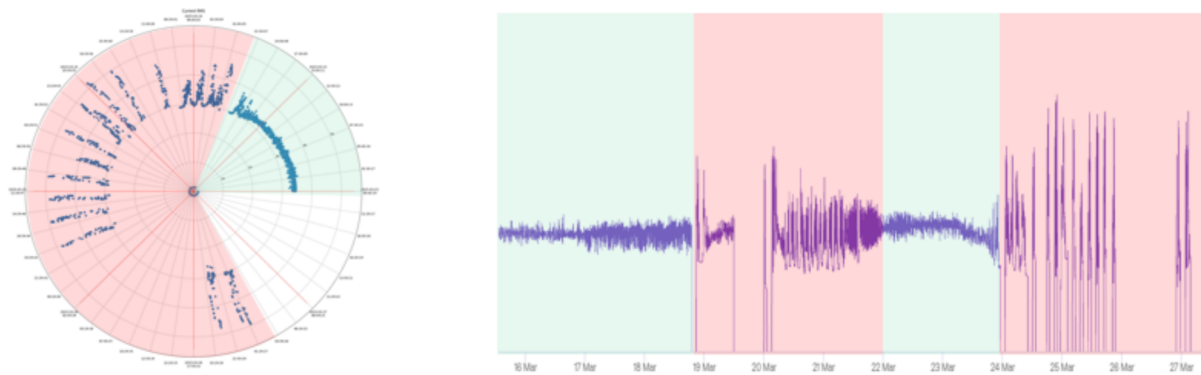


Image 5—Left: Ammeter chart based on CurrentRMS showing signs of excessive cycling; Right: Power Active over time, showing unstable operation and unexpected power spikes

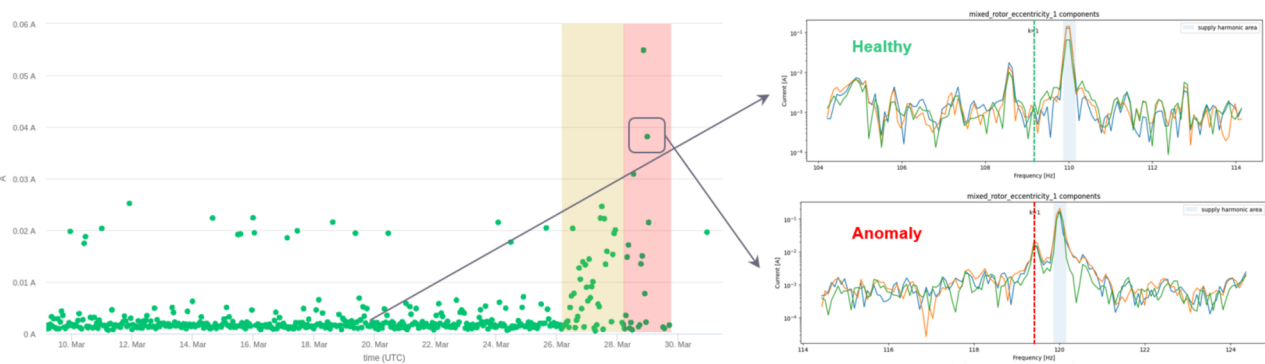


Image 6—Left: Rotational frequency component (mixed rotor eccentricity) tracked over time, showing an unexpected increase in energy; Right: Frequency spectra of two points with highlight the rotational frequency component within the spectrum, showing elevated energy (bottom) compared to the healthy sample (bottom)

Mechanical Unbalance. The SAM4 system detected increased energy around the 2nd and 4th harmonics visible only in the voltage and not the current, indicating mechanical imbalance and possible wear or on one of the rotors within the ESP. If the distortion were also visible in the current the cause is typically non-mechanical of nature.

After changing the operational mode by switching supply frequency more often causing the load in the ESP to fluctuate. The hypothesis was that this behavior could clear any debris within the ESP that caused the unbalance. After first a period of incidental frequency changes followed by a period of rapid changes the energy around the harmonics returned to a healthy level. It is suspected that dirt or soil caused a partial clogging or imbalance within the ESP, which was cleared after changes in the pump's RPM.

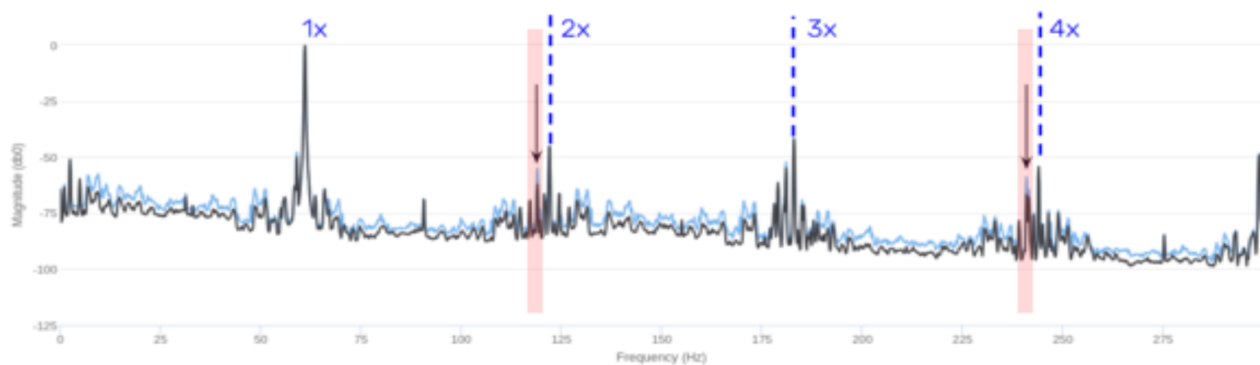


Image 7—Frequency spectra of two voltage measurements (healthy: black; unhealthy: blue) highlighting the increased energy around the 2nd and 4th harmonics

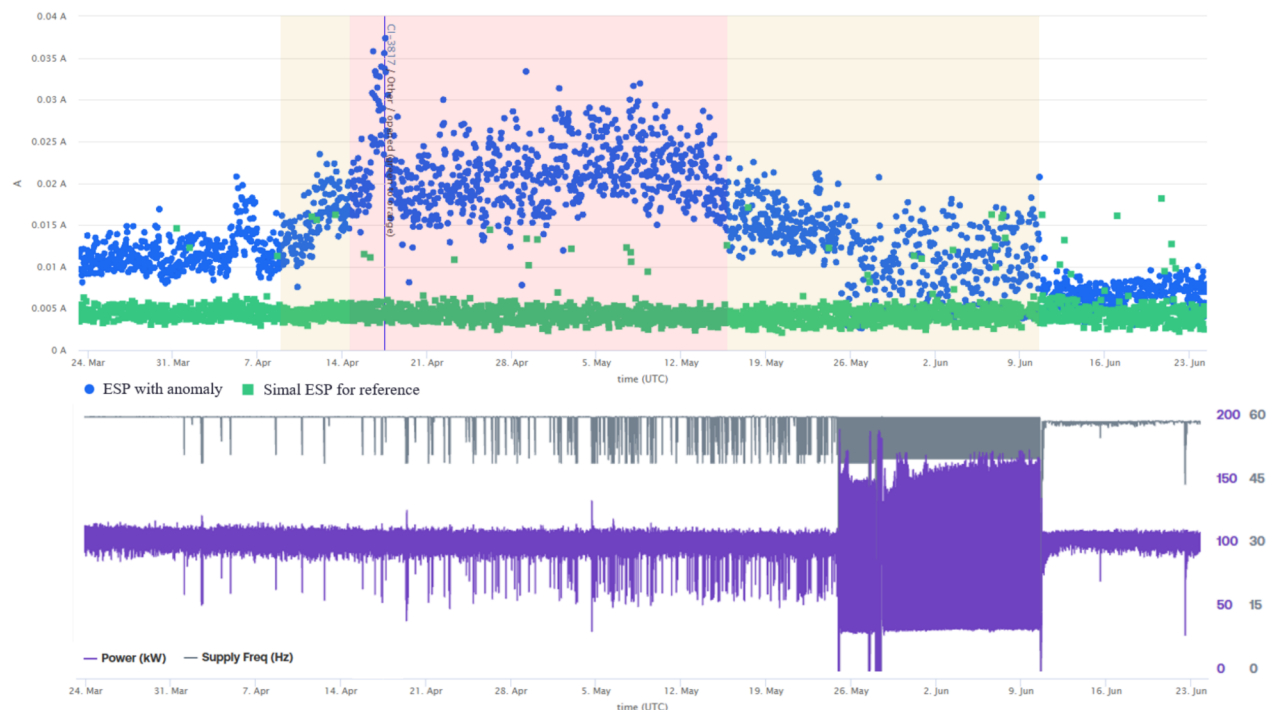


Image 8—Top: Energy of the 2nd harmonic tracked over time of the anomalous ESP (blue) and a reference (similar) ESP (green); Bottom: Supply frequency and active power within the same timeframe showing supply frequency changes resulting fluctuations in load

Learnings and Current Status

The field tests provided several key learnings:

- The integrity of the ESA signal is preserved despite the long cables and transformers, confirming the viability of surface-based monitoring.
- Existing centrifugal-pump algorithms require only minor adaptations for ESPs, accelerating the rollout of the technology.
- The high load variability of ESPs in unconventional wells necessitated the development of adaptive baseline logic.
- The importance of robust telemetry was highlighted, leading to the standardization of local buffering and the planned development of an edge mode.

As of Q2 2025, the SAM4-ESP system is commercially available and is being used by operators as either a primary monitoring system in wells without gauges or as a redundancy layer in high-volume wells.

Table 3—Key Learnings and Impact

Learning	Impact
Cable + transformer do not distort the signal in a way that ESA diagnostics are impossible.	Confirms surface viability.
Centrifugal-pump models port with minor tuning.	Accelerates rollout.
ESPs (in unconventional wells) show higher load variability.	Led to adaptive baseline logic.
Reliable telemetry is vital.	Local buffering now standard; edge mode planned.

Future Roadmap

The future development of the SAM4-ESP system is focused on several key areas:

Table 4—Future Development Roadmap

Item	Rationale
1 Friction-free install	Snap-fit harness, QR onboarding to cut technician time.
2 Coverage expansion	Identification of multiple components degradation, development of a compound-fault classifier.
3 Performance optimisation	Pump-curve overlay; PMM efficiency advisor for motor tuning.
4 Cloud-less / edge mode	Full local rule-engine, historian, and HTML5 dashboard for sites with intermittent connectivity—eliminates reliance on real-time cloud, with batch sync when available.
5 Ecosystem integration	REST/MQTT API, historian plug-in, SCADA widgets.
6 Adjacent assets	Geothermal ESPs, deep-well water pumps.

Broader Implications and Vision

The adoption of surface-based ESA for ESP monitoring has several broader implications for the oil and gas industry:

- **Cost Curve Shift:** Surface ESA, with its low data traffic requirements (≤ 2 GB/month), allows operators to monitor wells that could not previously justify the cost of downhole gauges. This "democratization" of proactive maintenance can lead to significant cost savings across the industry (Akselos, n.d.).
- **Enhanced Resilience:** The use of edge buffering, and soon edge analytics, will ensure continuous monitoring coverage even in areas with unreliable bandwidth, a critical factor for remote and offshore operations (IIoT World, 2025).
- **Improved Energy and Performance:** Early alerts from ESA systems can help reduce deferred production, minimize energy waste, and reduce the need for costly and potentially hazardous rig interventions (Nanoprecise, n.d.).
- **Path to Standardization:** The non-invasive nature, low capital expenditure, and low-bandwidth operation of ESA position it to become the default methodology for ESP health monitoring, setting a new industry standard.

Conclusion

The Permian Basin pilot project confirms that surface Electrical Signature Analysis delivers timely and accurate diagnostics for ESPs without the need for well intervention. With an installation cost an order of magnitude below that of downhole sensing, low-bandwidth operation (approximately 2 GB/month), and a clear roadmap toward a fully cloud-less edge version, the SAM4 system offers a practical and scalable solution for improving uptime and reducing lifting costs. ESA is poised to redefine condition monitoring for ESPs and other high-value assets that are hidden beyond the reach of conventional sensors, offering a more efficient, cost-effective, and resilient approach to asset management in the oil and gas industry.

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